Morgan Wang

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Technical Skills

Design and Analysis: Altium Designer, KiCad, LTspice, PCB Design, Microsoft 365, FDM 3D Printing
Hardware: Arduino UNO, ESP32, Soldering, Circuit Prototyping, Sensor Interfacing, Analog/Digital I/O, I2C, UDP/Wi-Fi
Languages and Tools: C++, Python, SQL, SQLite, Java, Git, GitHub, Arduino IDE, DigiKey

Projects

ESP32 Wireless Telemetry Prototype October 2026

ESP32, C++, Python, SQL, SQLite

- Built an ESP32 telemetry prototype for the UW Formula Electric TCU redesign, streaming sensor data over Wi-Fi via UDP at ~17.5 packets/s to a Python logger storing every packet in SQLite
- Ran 18 range tests (~8,300 packets) across 6 distance and obstruction conditions, using sequence numbers and SQL queries to measure packet loss, signal strength and dropout length
- Found loss stayed ≤ 0.4% at most positions up to 15 m, but reached 12% at one 10 m spot and 42.7% behind a closed door, with dropouts up to 5.8 s, showing the TCU needs a long-range radio and onboard buffering

5 V DC-DC Buck Converter September 2026 - October 2026

LTspice, Altium Designer, PCB Design

- Designed a 10–15V to 5V, 1A DC-DC buck converter using an LT1375 switching regulator, selecting a 15 µH inductor, Schottky freewheeling diode, input/output capacitors, and resistive feedback network through circuit analysis and datasheet calculations.
- Designed and routed a compact 41.4 × 23.6 mm two layer SMD PCB in Altium Designer, incorporating 25 mil power traces, 10 mil signal traces, a bottom layer GND polygon pour, and via connected ground returns.
- Optimized output capacitor ESR (0.10 Ω to 0.05 Ω) and feedback loop compensation (1 kΩ / 3.3 nF) in LTspice, reducing simulated output ripple from 45.7 mVpp to 23.6 mVpp (48.3%), achieving 32.7% lower ripple than the manufacturer's reference under matched 15V input and 1A load conditions.

Insulation Monitoring Device (IMD) Test Box - UW Formula Electric September 2026 - October 2026

Altium Designer, Safety Analysis, High Voltage Electronics

- Designed an IMD test box for UW Formula Electric, interfacing with the vehicle's HV+/HV− TSMP and chassis ground on Altium to simulate PASS, WARNING, TECH FAIL, and hard short conditions for FSAE high voltage safety verification.
- Developed the Altium Designer schematic and resistor networks, accounting for 15 kΩ TSMP protection resistance and calculating 285 kΩ, 485 kΩ, and 75 kΩ equivalent fault resistances to evaluate insulation monitoring thresholds.
- Analyzed high voltage insulation requirements for a 588V tractive system, calculating a 147 kΩ critical insulation resistance using the FSAE 250 Ω/V threshold to guide fault resistance selection and circuit design.

Morgan's Metronome v1.0 July 2026 - September 2026

Arduino UNO, C++, KiCad, PCB Design, Embedded Systems

- Designed and programmed an embedded digital metronome in C++ using an Arduino UNO, with adjustable 40–180 BPM, potentiometer control, push button inputs, and a 16×2 LCD.
- Implemented non blocking timing with millis() to maintain beat intervals while keeping tempo and button inputs responsive.
- Prototyped and debugged the circuit on a breadboard before transferring it into an Arduino UNO compatible shield on KiCad.

Experience

Electrical and Firmware Team Member September 2026 - Present

University of Waterloo Formula Electric *Waterloo, Ontario*

- Collaborating with electrical and firmware engineering teammates in a multidisciplinary Formula SAE design team, contributing to circuit design and technical evaluations of vehicle electrical systems using Altium and LTSpice.
- Contributing to the redesign of the vehicle's Telematics Communication Unit by reviewing the existing cellular modem/SIM based architecture and investigating alternative wireless technologies for more reliable, lower cost real time telemetry and post run data analysis.

Badminton Program Instructor August 2025 - May 2026

City of Richmond Hill - Rouge Woods Community Centre *Richmond Hill, ON*

- Coached beginner and intermediate badminton classes of 15 students, tailoring lesson plans to individual student needs.

Education

University of Waterloo September 2026 - April 2031

Bachelor of Applied Science, Honours Electrical Engineering (Co-op) *Waterloo, ON*